

Quiz-1

Numerical

- 1- Choose the correct order
 - a. Removal of stopword, stemming, tokenization
 - b. Tokenization, removal of stopword, stemming
 - c. Tokenization, stemming, removal of stopwords
 - d. Removal of stopword, tokenization, stemming
- 2- Calculate the precision of the IR System if the total available documents are 95 in which 42 are the relevant documents and 34 are irrelevant documents
 - a. 0.55
 - b. 0.40
 - c. 0.62
 - d. 0.50
- 3- The disadvantage of Boolean retrieval model is
 - a. Easy to implement
 - b. Ranking is difficult
 - c. Query processing is difficult
 - d. A complex retrieval model
- 4- To improve the recall of an information retrieval system by adding related terms to the query the technique used is/are
 - a. Query expansion
 - b. Relevance feedback
 - c. Pseudo relevant feedback
 - d. None
- 5- An inverted index
 - a. A database that stores, for each term t , the list of all documents that contain term t
 - b. Stores mapping from documents to words
 - c. Orders the term in a different order which is not a sequential order
 - d. All of the above
- 6- Which of the following is/are true about TF-IDF:
 - a. TF-IDF is best known weighting scheme in information retrieval
 - b. TF-IDF increases with the number of occurrences of term within a document
 - c. TF-IDF increase with the rarity of the term in the collection
 - d. A metric derived by taking the log of N divided by the document frequency where N is the total number of documents in a collection is called Document Frequency
- 7- A crude heuristic process that chops off the ends of the words to reduce inflectional forms of words and reduce the size of the vocabulary is called:
 - a. Lemmatization
 - b. Case folding
 - c. Stemming

- d. None of the above
- 8- One disadvantage, as outlined in our text, of using a permuterm index for wild card queries is:
- a. It requires complex code that is difficult to maintain
 - b. It has the risk of key collisions which are difficult to resolve
 - c. The required rotations creates a very large dictionary
 - d. It cannot be used to find terms that are not spelled correctly
- 9- Which of the following is a technique for context sensitive spelling correction:
- a. the Jaccard Coefficient
 - b. Soundex algorithms
 - c. k-gram indexes
 - d. Levenshtein distance
- 10- The purpose of the inverse document frequency is to increase the weight of terms with high collection frequency.
- a. True
 - b. False
- 11- In information retrieval, extremely common words which would appear to be of little value in helping select documents that are excluded from the index vocabulary are called:
Select one:
- a. Stop Words
 - b. Tokens
 - c. Lemmatized Words
 - d. Stemmed Terms
- 12- What is the primary purpose of a web crawler for search engines?
- a. To improve website design and layout
 - b. To encrypt and secure website data
 - c. To gather information about websites to improve search results
 - d. To analyze website traffic and user behavior
- 13- Which of the following statements about Bloom filters are correct? ☐
- a. A Bloom filter has the same properties as a standard Hashmap data structure.
 - b. A Bloom filter is full if no more hash functions can be added to it.
 - c. A Bloom filter always returns FALSE when testing for an element that was not previously added.
 - d. A Bloom filter always returns TRUE when testing for a previously added element.
 - e. An empty Bloom filter (no elements added to it) will always return FALSE when testing for an element.
- 14- Bloom filter guarantees no
- a. false positives

- b. false negatives
- c. false positives and false negatives
- d. false positives or false negatives, depending on the Bloom filter type

15- How does a web crawler determine which pages to visit on a website?

- a. By following links on the website's homepage
- b. By randomly selecting pages on the website
- c. By using keywords to search for relevant content
- d. By visiting every page on the website

16- Consider the strings “monday” and “tuesday”. What is the edit distance between the two strings?

- a) 3
- b) 4
- c) 5
- d) 6

17- A data structure that maps term back to the parts of a document in which they occur is called an

- a. Posting list
- b. Incidence matrix
- c. Dictionary
- d. Inverted index

18- Edit distance is _____ type of spelling correction

- a. Isolated term correction
- b. Context sensitive correction
- c. K-gram correction
- d. Both A and B

19- _____ is fraction of returned result relevant to information need

- a. Precision
- b. Recall
- c. Corpus
- d. F-measure
- e. Relevance

20- Boolean retrieval model does not provide provision for:

- a. Ranked search
- b. Proximity search
- c. Phrase search
- d. Both proximity and ranked search

21- Benefits of using B-trees:

- a. Re-balancing is cheap
- b. Balanced trees allow efficient retrieval
- c. Faster $O(\log M)$
- d. Solves the prefix problem.